Beyond Tenant Isolation

A Flow-Level Testbed for Common-Mode Resilience in Cloud-Dependent Backhaul

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Abstract

Cloud outages rarely respect tenant boundaries. A provider control plane, identity service or AI endpoint fails once and every deployment behind it fails together. Current cloud isolation — logical tenants, confidential virtual machines, regions and provider diversity — cuts tenant-to-tenant, workload-to-host and region-to-region, but not the shared provider control plane. Multipath is often proposed as the complement, but paths that share a conduit or a control plane are one path with extra steps. We present a flow-level testbed that compares nine architectures under ten outage events with 30 paired seeds and 2,700 runs, with the hypotheses and thresholds frozen in version control before the simulator existed. The architectures range from a single provider to tenant isolation, SRLG-disjoint multipath, independent control planes, a one-way boundary with a tested fallback, an untested fallback and a global cooperative controller.

Under the frozen scenario the model makes five checks, not empirical claims. Tenant isolation contains a tenant-shared gateway fault (100% vs 0%) but not an identity failure (0% vs 0%); three paths over one conduit fail together (0% vs 100% for SRLG-disjoint paths); a one-way boundary with a tested fallback holds 90% under an AI control-plane failure against 0% without it; the isolation-multipath interaction is +44.4 points; and a global controller loses 100 points when it fails with its primary path. A tested fallback is worth 34.5 points over an untested one on the same failure. Under the stipulated model the checks bear on three claims: current cloud isolation is the wrong cut for common-mode failure; multipath must be independent, not merely multiple; and a controller is a dependency, not an abstraction above dependencies.

Every one of those numbers is conditional on a stipulated constant, so the paper also reports a one-factor sensitivity analysis and break-even values. H18 fails only below 0.85 local-policy efficiency (stipulated 0.90); the tested-fallback gap closes below 0.22 activation probability (stipulated 0.98), above 18.2 s activation delay (2 s), or below 0.122 Gbps fallback capacity (0.50 Gbps); the untested fallback would have to improve beyond the swept range to erase the gap. H19's absolute level fails below 0.80 efficiency, while its interaction term alone survives to 0.656. Evidence is a stipulated flow-level scenario, not a production measurement. The break-even table is the honest form of the result: it shows how far each conclusion is from its stipulated operating point, and the paper states what would count as a refutation.

Keywords: common-mode failure; cloud resilience; network isolation; multipath; data diode; tested fallback; SRLG; cooperative backhaul.

1. Introduction

In the same week of November 2026 two unrelated disruptions were reported. On 13 November the FAA issued a nationwide ground stop after an old telecommunications system used by air-traffic control was overloaded; the stop lasted about four hours and twenty minutes, and the FAA and the Department of Homeland Security reported no evidence of a cyber or AI attack [15,16]. Separately, Microsoft reported a Microsoft 365 outage that it attributed to an issue introduced during an earlier network capacity upgrade; Outlook, Teams, Copilot and other services were affected, and Microsoft said a full post-incident report would follow [17,18]. AI services including ChatGPT and Claude also reported disruption during the same period [18]. These are separate events with separate triggers; the public reports do not establish a common cause.

The same kind of coincidence appears earlier. On 2-3 September 2026 Microsoft 365 reported degradation affecting Outlook, Teams, SharePoint and Copilot [19], while on the evening of 3 September ChatGPT, Claude and Grok reported elevated errors [20,21]. The AI providers described infrastructure and compute-center problems; the public reports do not establish that the Microsoft 365 degradation and the AI outages shared a root cause. The coincidence is a reason to ask the structural question, not evidence for the answer.

The structural cause is concentration. A dependency shared by many consumers, with no independent failure domain behind it, turns a local fault into a correlated outage. Let a service depend on a shared node d and an independent part i. Its availability is bounded by

$$A_{\text{service}} \;\leq\; A_d \cdot A_i.$$

Adding consumers to d increases scale, utilization and blast radius at the same time. It does not add an independent failure domain; it widens the one that already exists. This is why concentration is economically rational and structurally dangerous: the same decision that lowers unit cost raises correlated exposure.

Cloud architecture does have isolation, but it isolates the wrong boundary for this failure. Logical tenant isolation (VPCs, IAM, hypervisors) cuts tenant-to-tenant; confidential computing and TEEs cut workload-to-host [30]; regions and availability zones cut region-to-region; provider diversity cuts provider A from provider B [1-4]. None of these cuts tenant-to-provider, consumer-to-control-plane, or consumer-to-shared-dependency. The weak point was never tenant-to-tenant. The shared control plane was.

The engineering response has two halves, and they are not substitutes. **Multipath and diversity** keep service running while a path, link or provider degrades. **Isolation** stops a fault from propagating across a boundary at all. Multipath cannot cut a shared identity service; isolation cannot create capacity on the protected side. A tested fallback is the condition that makes either one real: an isolation boundary that leaves the protected side without capacity converts a provider outage into a local outage, and an unexercised fallback is a control arm, not a mitigation.

This paper turns that argument into a testbed and a comparison. We do not propose a new routing protocol. We ask a narrower, falsifiable question: under a common-mode event, which architectural choices actually preserve service, and how do they compare with the isolation categories current cloud architecture already provides?

Contributions.

1. A flow-level testbed with nine architectures, ten outage events, 30 paired seeds and 2,700 runs, with the hypotheses and thresholds frozen at commit `1c781f3` before the simulator existed.
2. A comparison between stylized current-cloud architectures — single provider, multi-zone and multi-region paths, tenant isolation, independent control planes and a global orchestrator — and the two-axis principles of isolation plus independent multipath plus a tested fallback.
3. Five model checks with thresholds frozen before the simulator existed: tenant isolation is the wrong cut for a shared control plane; multipath gain tracks path independence, not path count; a one-way boundary without local capacity is a partition; the isolation- multipath interaction is positive; and a global cooperative controller is a new common-mode carrier.
4. An open artifact: frozen protocol, simulator, benchmark, a one-factor sensitivity and break-even analysis, figures, raw CSV outputs and tests.

What we do not claim. This is not a measurement of AWS, Azure, Google Cloud, Cloudflare or any production network. The current-cloud architectures in Table 1 are structural analogues, not reconstructions of a particular provider. The model is flow-level: it computes capacity and dependency conjunction, not packets, queues, TCP behaviour or hardware. Security is not evaluated. The motivating outages are reported events; their root causes are not independently verified here, and no result depends on a particular root cause being correct. No result is empirical support for the model's claims; the model checks are statements about the model, and the postmortem study is the separate empirical test.

2. Related Work

2.1 Failure domains and isolation

Cloud computing is defined by on-demand, pooled resources and rapid elasticity [1], and the major providers expose isolation through regions and availability zones [2-4]. Tenant isolation is the default boundary: a tenant's virtual network, identity policy and hypervisor namespace are separate from another tenant's. Confidential computing extends this to the host, placing a workload in a hardware-enforced trusted execution environment. These mechanisms are effective for the threat they address. They do not address a provider control-plane fault, because the control plane is outside the tenant boundary by construction.

Dependability taxonomy separates faults, errors and failures, and distinguishes independent from common-cause failures [5]. Cloud measurement studies document that many cloud incidents are configuration and software faults rather than hardware faults [6]. Chaos engineering formalises the practice of testing failure paths before they are needed [7]. Operational technology security guidance recommends segmentation at the IT/OT boundary [8], and data diodes provide hardware-enforced one-way transfer at that boundary [9]. Rail signalling provides a vendor-described existence proof of a data diode paired with commercial off-the-shelf servers [9]. We treat the data diode as an availability and attack-surface boundary, not as a cryptographic control.

2.2 Multipath and independence

Multipath transport is standardised in MPTCP [10], and equal-cost multipath routing is standard in datacentre fabrics [11]. Shared-risk link groups (SRLGs) are the conventional way to express that two links are not independent [12]. The multipath literature usually evaluates throughput, latency and failover; it less often asks whether the path set is independent of the *control* plane that selects it. A three-path set that reconverges on one identity service or one controller is a single failure domain with three next hops.

Cooperative backhaul adds a second form of concentration. Self-scaling pools capacity across links and tenants, which raises utilization, but a global cooperative controller is itself a shared dependency. The controller can become the common-mode carrier that the multipath was meant to remove. We make controller locality an explicit design variable rather than an implementation detail.

2.3 Testbeds and pre-registration

Packet-level testbeds are the right instrument for protocol behaviour. They are a poor instrument for a structural question that spans many architectures and outage types: the cost of building each architecture in a kernel is high, and the result is still tied to one kernel and one topology. Our earlier layering testbed followed the opposite route, using a model, a discrete-event simulation and a Linux namespace harness to separate structural from implementation effects [13]. The present study is deliberately flow-level: it isolates the dependency structure and the path-independence condition, and leaves packet-level behaviour to a later fidelity. Paper B in the same repository is a caution against assuming that a biologically inspired controller will beat a simpler structural mechanism [14]; the biological parallel in this paper is a hypothesis generator, not evidence.

2.4 Positioning

Prior work treats isolation and multipath as separate remedies: isolation is a tenancy or segmentation property, multipath is a transport or routing property. The claim tested here is that they are independent mitigations of the same failure, and that neither substitutes for the other. The comparison is against stylized current-cloud architectures, not against a new protocol. The contribution is the joint test, the frozen protocol and the measured interaction, not a claim of priority over the individual mechanisms.

2.5 Empirical outage studies and provider postmortems

The empirical literature on cloud outages is thinner than the operational literature, but it is not absent. Gunawi et al. coded more than 3,000 issues in cloud systems [6] and later asked why the cloud stops computing in a study of hundreds of service outages [22]. Talluri et al. release the Cloud Uptime Archive, a multi-vantage availability dataset, and report mean time between failures, recovery times and operator-stated root causes; in their one-year operator subset, configuration errors dominate multi-regional outages [23]. Paladugu coded 70 official postmortems from 22 providers (2019-2026): changes started 52 of the 70 incidents (74.3%), configuration changes alone accounted for 28 of the 52 change-triggered incidents (53.8%), and 10 of 11 globally pushed changes produced wide impact [24]. That study is a preprint, not peer-reviewed, and its frame is published postmortems, not all outages.

The provider corpora are the primary material: AWS Post-Event Summaries [25], Azure post-incident reviews [26], Google Cloud Service Health [27] and Cloudflare's postmortem archive [28]. Their disclosure depth and cadence differ, so they cannot be treated as a random sample. Two Cloudflare examples show the class this paper models: a December 2025 configuration-change outage lasted about 25 minutes, and a November 2025 bug in a Bot Management feature file affected many services [28]. We do not code those incidents here; the companion postmortem study does that under a frozen codebook, and its results are reported separately.

3. Principles and Model

3.1 Six principles

The testbed encodes six principles. They are stated before the results because they determine what the model measures.

1. **Availability is a conjunction.** A tenant is served only if its data path, its control plane, its tenant boundary and, where required, its fallback are available at the same time.
2. **Multipath counts only across independent failure domains.** Paths that share a conduit, a provider control plane or a shared control service are one path with extra steps.
3. **Isolation trades capability for containment.** A cut that removes a shared dependency can also remove the capability that dependency provided. A boundary without local capacity is a partition, not a mitigation.
4. **A fallback is a tested state.** It has an activation probability, a delay and a capacity fraction. An untested fallback is a control arm.
5. **The controller is a dependency.** Self-scaling through a global cooperative controller concentrates the service it is meant to diversify.
6. **Common-mode events are sampled once per event, not per tenant.** A shared node fails for everyone at once; that is what makes it common-mode.

3.2 Availability model

A deployment is a dependency graph $G = (V, E)$. A path set P serves a tenant, with path capacity c_p, latency and an SRLG set. Two paths are independent when their SRLG sets are disjoint and their providers differ. The **path independence index** is

$$I(P) \;=\; \frac{\max\{|S| : S \subseteq P,\ S \text{ pairwise independent}\}}{|P|}.$$

Under independent path failures with availabilities a_p, the path-set availability is

$$A(P) \;=\; 1 - \prod_{p \in P} (1 - a_p),$$

but under a common-mode event the same expression is capped by the shared dependency's availability. The cap, not the product, is the object of interest here.

A tenant also requires its control plane. If the shared identity or DNS service is down, the path set is unusable even when every link is up; if the AI endpoint is on the availability path, an AI failure is a service failure. An isolation boundary is a cut C in G. When the cut removes a required dependency, the tenant serves at most its local fallback capacity c_{fb}:

$$A_{\text{iso}} \;=\; \min\left(1, \frac{c_{fb}}{D}\right),$$

where D is demand. A fallback with activation probability q and delay d contributes $q{\cdot}\min(1, c_{fb}/D)$ only after d; the delay is averaged into the event window. A global controller that fails reduces the usable path set to the primary path; a local controller does not.

3.3 Architectures

Table 1 lists the nine arms. The first five and A8 are stylized versions of current-cloud choices; A6 instantiates the proposed combination; A7 is a fallback-quality control; A9 is the isolation-only control.

Table 1. Architectures.

Arm	Paths	Tenant iso.	Control	One-way	Fallback	Controller
A1	1 shortest	no	shared	no	none	none
A2	3 shortest	no	shared	no	none	local
A3	3 disjoint	no	shared	no	none	local
A4	3 disjoint	yes	shared	no	none	local
A5	3 disjoint	yes	independent	no	none	local
A6	3 disjoint	yes	independent	yes	tested	local
A7	3 disjoint	yes	independent	yes	untested	local
A8	3 disjoint	yes	independent	no	none	global
A9	1 shortest	yes	independent	yes	tested	none

Shared control means the tenant requires global identity, DNS and the AI endpoint. **Independent control** means identity and DNS are local; the AI endpoint is still required unless the arm is one-way. **One-way** removes the AI endpoint from the availability path and serves path traffic at 0.9 efficiency (the stipulated cost of local policy). **Tested fallback** activates with probability 0.98 after 2 s at 0.50 Gbps; **untested** activates with probability 0.60 after 8 s at 0.30 Gbps. **Global controller** uses only the primary path when the controller is down; **local** reallocates across all surviving paths. These constants are the ones varied one at a time in Section 5.9; the topology, event targets, demand jitter and seed set stay fixed.

3.4 Events

Table 2. Outage events.

Event	Failure	What it tests
`none`	no failure	normal operation
`link`	`a1-p1` down	one path, independent failure
`srlg`	all `a1-p*` links down	shared conduit
`backhaul_cut`	all agg-to-provider links down	total path loss, fallback
`provider_cp`	`p1-cp` down	provider control plane
`idp`	global identity down	shared identity
`ai`	AI endpoint down	shared AI control plane
`tenant_gw`	shared tenant gateway down	tenant-shared fault
`controller_link`	controller and `a1-p1` down	controller locality
`ai_link`	AI endpoint and `a1-p1` down	combined control and path failure

All events start at $15 + U(0, 10)$ seconds and last $20 + U(0, 10)$ seconds in a 60 s run. The start and duration are drawn once per seed and shared by all arms.

3.5 Mapping to current cloud architecture

The mapping is structural, not a claim about a specific provider.

A1 is a single-region, single-provider deployment: one path, one control plane, no fallback. **A2** is a multi-zone or multi-region path set that shares a conduit and a global control plane: more next hops, one failure domain. **A3** adds SRLG-disjoint paths, which is the strongest path diversity current cloud offers, but keeps global identity, DNS and AI. **A4** adds tenant isolation, the default cloud boundary. **A5** adds independent identity and DNS, the strongest single-tenant control-plane design in the set, but the AI endpoint remains on the availability path. **A8** is the global orchestrator: a single cooperative controller that can reallocate across all paths and tenants.

A6 is the proposed architecture: disjoint paths, tenant isolation, independent control, a one-way boundary at the AI/control interface, a tested local fallback and local reallocation. **A9** isolates without multipath: the same boundary and fallback on a single path. A9 is the control that separates "isolation works" from "isolation is sufficient".

4. Testbed and Protocol

4.1 Scenario

Six tenants connect through three aggregation sites to three providers and two cloud regions. Each tenant has three edge-to-aggregation links (1.05 Gbps, 1.0-1.5 ms), each aggregation site has three provider links (1.20 Gbps, 4.0-7.0 ms), and provider-to-cloud links run at 1.50 Gbps (8.0-9.0 ms). All provider links from one aggregation site share an SRLG (agg-j), so one conduit failure takes all of that site's provider links. The path bottleneck is 1.05 Gbps; mean demand is 1.0 Gbps with a per-seed 5% jitter. A path is edge $\rightarrow$ aggregation $\rightarrow$ provider $\rightarrow$ cloud; its capacity is the minimum link capacity and its SRLG set is the union of link SRLGs.

Shared dependencies are global identity (`idp`), DNS (`dns`), the AI endpoint (`ai`), the tenant gateway (`mt-gw`) and the cooperative controller (`coop`). Each tenant has a local fallback. The scenario is a testbed for the structural hypotheses, not a model of a specific operator.

4.2 Metrics

For each arm, event and seed we record:

- **availability during the event**: mean served/offered over tenants and event ticks;
- **overall availability**: mean over the 60 s run;
- **recovery**: time from event start until a tenant first reaches 90% for three consecutive ticks, or the event duration if it never does;
- **blast radius**: fraction of tenants below 50% during the event;
- **path independence**: the index of Section 3.2;
- **fallback rate**: fraction of tenants whose fallback activated;
- **cost index**: 0.5 per path, plus 0.1 each for tenant isolation, independent control and one-way, plus 0.05 for a global controller.

The cost index is a stipulated accounting device, not a price. It makes the trade-off visible; it is not used to rank architectures.

4.3 Protocol and reproducibility

The hypotheses and thresholds are frozen in common-mode-predictions.md at commit 1c781f3. The benchmark runs 30 seeds (base 2026), nine arms and ten events: 2,700 runs. Each seed draws one event, one demand vector and one fallback vector; every arm sees the same draws. The simulator is deterministic given a seed. The benchmark writes per-seed, summary and verdict CSVs under output/common-mode/. The full reproduction command is

```
py -3 tools/benchmark_common_mode.py --seeds 30
py -3 tools/benchmark_common_mode.py --sensitivity --seeds 30 --sensitivity-points 25
py -3 -m unittest tests.test_common_mode tests.test_common_mode_sensitivity -v
py -3 tools/make_common_mode_figures.py
```

The benchmark writes per-seed, summary and verdict CSVs under output/common-mode/. The sensitivity mode writes common-mode-sensitivity.csv, common-mode-break-even.csv and a generated Markdown table under docs/paper-tables/; the paper pastes that table rather than retyping numbers. The figures in this paper are generated by tools/make_common_mode_figures.py from the committed summary, per-seed, sensitivity and break-even CSVs. The full-suite count and environment are recorded in TASKS-STATUS.md and the release audit.

5. Results

5.1 Overall picture

Figure 1 shows availability for every arm and event. The pattern is not a single ranking. A1 is available only with no failure. A2 survives an independent link failure but not a conduit failure. A3 survives both path events but not identity or AI. A4 adds tenant isolation and survives the tenant gateway. A5 adds independent identity and survives identity, but not AI. A6 survives AI, link, conduit, provider control, identity and tenant faults, and serves 45.6% under a full backhaul cut. A8 behaves like A5 until its controller fails, then falls to 0%. A9 matches A6 on the common-mode events but is much weaker under an independent link failure.

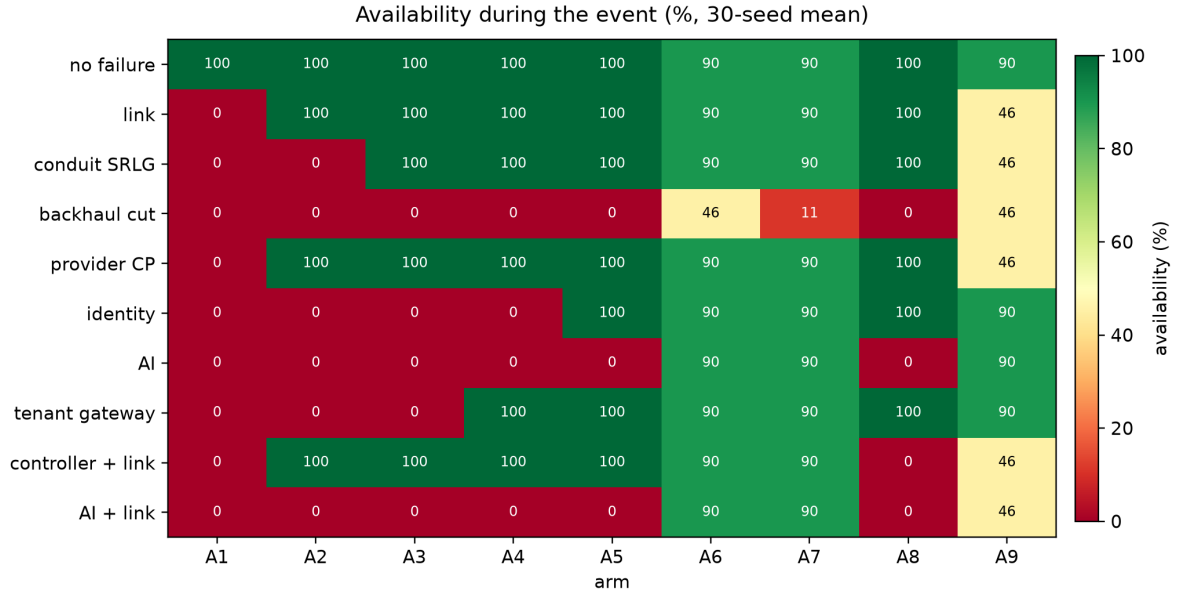


Figure 1. Availability during the event for the nine architectures and ten outage events (30-seed

mean). Values are percentages; the color scale is shared. Data: `docs/paper-tables/common-mode-summary.csv`, regenerated by `tools/make_common_mode_figures.py`.

5.2 H16: tenant isolation is the wrong cut

Tenant isolation turns the tenant-shared gateway failure from a total outage into normal service: A3 is 0% and A4 is 100%. It does nothing for the identity failure: A3 and A4 are both 0%. The result is not that tenant isolation is useless; it is that the cut is in the right graph for tenant-to-tenant faults and in the wrong graph for the shared control plane. The shared node lies outside the tenant boundary, so no amount of tenant separation reaches it.

This is the category error in its cleanest form. Current cloud isolation is designed to prevent one tenant from reaching another; it is not designed to keep a tenant serving when the provider's identity or AI service is down. The fault that matters here crosses the boundary that tenant isolation does not cut.

5.3 H17: independence, not count

A2 and A3 both have three paths. Under the `agg-1` conduit failure A2 is 0%, because all three of its shortest paths share the conduit, while A3 is 100%, because its paths use three aggregation sites. Under the independent `a1-p1` link failure A2 is 100% and A1 is 0%. Path count alone would have predicted the opposite ordering for the conduit event.

The path independence index makes the difference explicit: A2 is 0.333 and A3 is 1.0. The same pattern appears for a provider control-plane failure: A1 is 0% and A2 is 100%, because A2 has paths via other providers. Multipath is a diversity property, not a multiplicity property. A path set that shares an SRLG or a control plane is a single failure domain with extra next hops.

5.4 H18: a boundary without capacity is a partition

Under the AI control-plane failure A5 is 0% and A6 is 90%. A5 has independent identity and DNS, but it still requires the AI endpoint; A6 removes that dependency with a one-way boundary and serves local policy at 0.9 efficiency. The normal-operation cost is 10 points: A5 is 100% and A6 is 90% with no failure. The cost is the price of local policy, not a failure of the boundary.

The fallback quality result is sharper. Under the full backhaul cut A6 serves 45.6% with a 99.4% activation rate; A7 serves 11.1% with a 56.1% activation rate. The tested fallback is worth 34.5 points on the same failure. An untested fallback is not a mitigation; it is a control arm that measures how often the organisation happened to configure the fallback correctly.

5.5 H19: the interaction is positive

Under the combined AI + link failure the four relevant arms are A1 = 0, A3 = 0, A9 = 45.6 and A6 = 90.0. The interaction term is

$$(A6 - A3) - (A9 - A1) = (90.0 - 0) - (45.6 - 0) = 44.38 \text{ points.}$$

The reading is direct. A3 has multipath but not isolation, so the AI failure takes it to 0. A9 has isolation but not multipath, so the link failure leaves it with fallback capacity only. A6 has

both, and reaches 90. Isolation and multipath are not substitutes; under a mixed failure they are complements.

5.6 H20: the controller is a dependency

Under the controller + primary-link failure A5 (local reallocation) is 100% and A8 (global controller) is 0%. With no failure both are 100%, so the global controller buys nothing in normal operation and costs everything when it fails: once it is down, A8 falls back to its primary path, which is the failed link. This is the cooperative-controller trap. Self-scaling through a global controller raises utilization and recreates the concentration the multipath was meant to remove. Local, partition-tolerant reallocation avoids it.

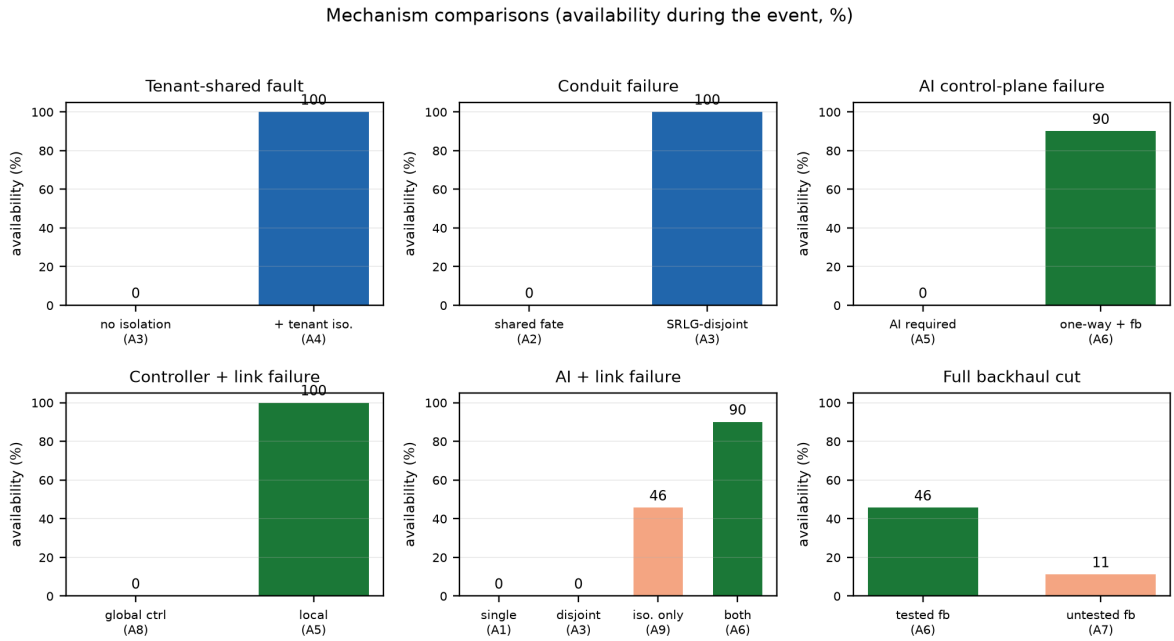


Figure 2. Mechanism comparisons. Each panel isolates one architectural choice: tenant isolation, path independence, the one-way boundary with fallback, controller locality, the combined interaction and fallback quality. Values are availability during the event (%); data as in Figure 1.

5.7 Comparison with current cloud architectures

Table 3 compares the stylized current-cloud arms with the proposed combination on the events that matter for common-mode failure. The table is the paper's central comparison. Arm identifiers follow Table 1: A1-A5 and A8 are the current-cloud analogues, A6 is the proposed combination, A7 is the untested-fallback control and A9 is the isolation-only control.

Table 3. Current-cloud analogues versus the proposed combination (availability during the event, %).

Arm	Normal	Tenant gw.	Identity	AI	AI + link	Cut	Cost
A1	100.0	0.0	0.0	0.0	0.0	0.0	0.50
A2	100.0	0.0	0.0	0.0	0.0	0.0	1.50
A3	100.0	0.0	0.0	0.0	0.0	0.0	1.50

Arm	Normal	Tenant gw.	Identity	AI	AI + link	Cut	Cost
A4	100.0	100.0	0.0	0.0	0.0	0.0	1.60
A5	100.0	100.0	100.0	0.0	0.0	0.0	1.70
A6	90.0	90.0	90.0	90.0	90.0	45.6	1.80
A7	90.0	90.0	90.0	90.0	90.0	11.1	1.80
A8	100.0	100.0	100.0	0.0	0.0	0.0	1.75
A9	90.0	90.0	90.0	90.0	45.6	45.6	0.80

Current cloud improves along one axis at a time. Tenant isolation (A4) fixes the tenant gateway but not identity or AI. Disjoint multipath (A3) fixes the conduit failure but not the control plane. Independent control (A5) fixes identity but not AI. The global controller (A8) does not fix any common-mode event on its own, and it adds a new one. No current-cloud arm reaches 90% under AI + link; A6 does.

The comparison also exposes a trade that a single ranking would hide. A9 matches A6 on the four common-mode events and costs less (0.80 vs 1.80), because both use the same one-way boundary and tested fallback. A6's extra cost buys resilience to the independent link failure (90% vs 45.6%). If the operator's failure model is common-mode only, isolation plus a tested fallback is a defensible architecture; if independent path failures are in scope, multipath is still required. The "both axes" claim is therefore conditional on the failure mix, not a claim that the most expensive arm dominates everywhere.

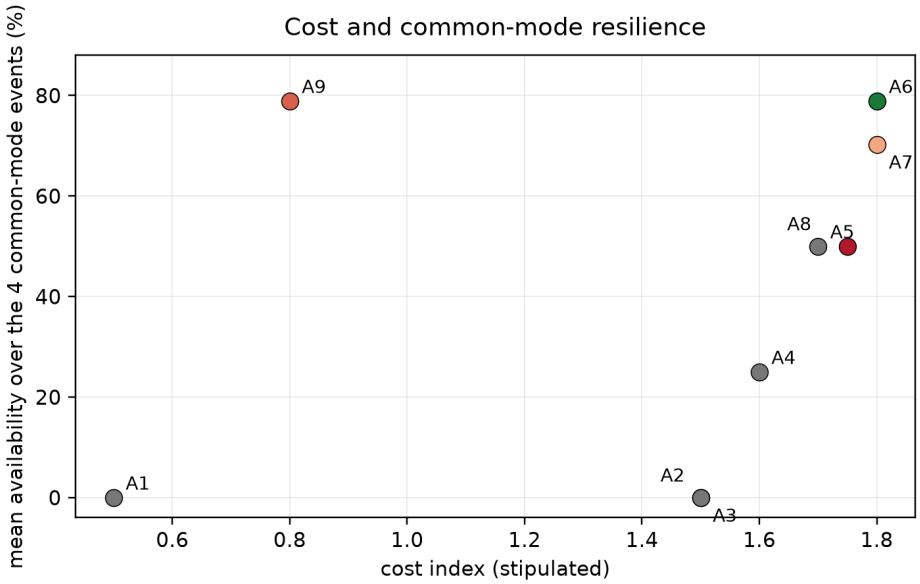


Figure 3. Cost index against mean availability over the four common-mode events (identity, AI, tenant gateway, backhaul cut). A9 matches A6 on this metric at a lower cost; A6 buys independent-link resilience (Section 5.7). The cost index is stipulated and is not a price.

5.8 Verdicts

Table 4. Frozen hypotheses and model checks (30-seed means).

Hypothesis and criterion	Value	Threshold	Verdict
H16 tenant isolation: A4-A3 tenant gateway; abs(A4-A3) identity	100.0; 0.0	>= 50; <= 5	PASS
H17 independence: A3-A2 SRLG; A2-A1 link	100.0; 100.0	>= 20; >= 50	PASS
H18 tested fallback: A6-A5 AI; A6 AI; A5-A6 normal	90.0; 90.0; 10.0	>= 50; >= 80; <= 15	PASS
H19 interaction: (A6-A3)-(A9-A1) AI + link	44.38	>= 20	PASS
H20 controller locality: A5-A8 controller + link; abs(A5-A8) normal	100.0; 0.0	>= 50; <= 5	PASS

Every criterion passes on the 30-seed mean and on every individual seed (pass fraction 1.000). The per-seed pass fractions and all twelve sub-criteria are in `docs/paper-tables/common-mode-verdicts.csv`. The all-pass result is a property of the structural model: the mechanisms are isolated by design, so the margins are large. It is not evidence that a production deployment will see the same numbers.

5.9 Per-seed distributions

Figure 4 shows every paired difference over all 30 seeds, not only the mean. The mechanisms that are deterministic in the model are flat: tenant isolation adds 100 points under the tenant-gateway failure on every seed, SRLG-disjoint multipath adds 100 points under the conduit failure, and local reallocation adds 100 points under the controller-plus-link failure. The fallback comparisons are not flat: the tested-minus-untested gap ranges over roughly 25-42 points, and the H19 interaction ranges over roughly 42-51 points, because the fallback draw and the event window vary by seed. In every one of the 30 seeds each frozen criterion passes; the per-seed pass fraction is 1.000. The figure makes the variance visible where it exists and shows that the deterministic comparisons are exact properties of the model, not lucky draws.

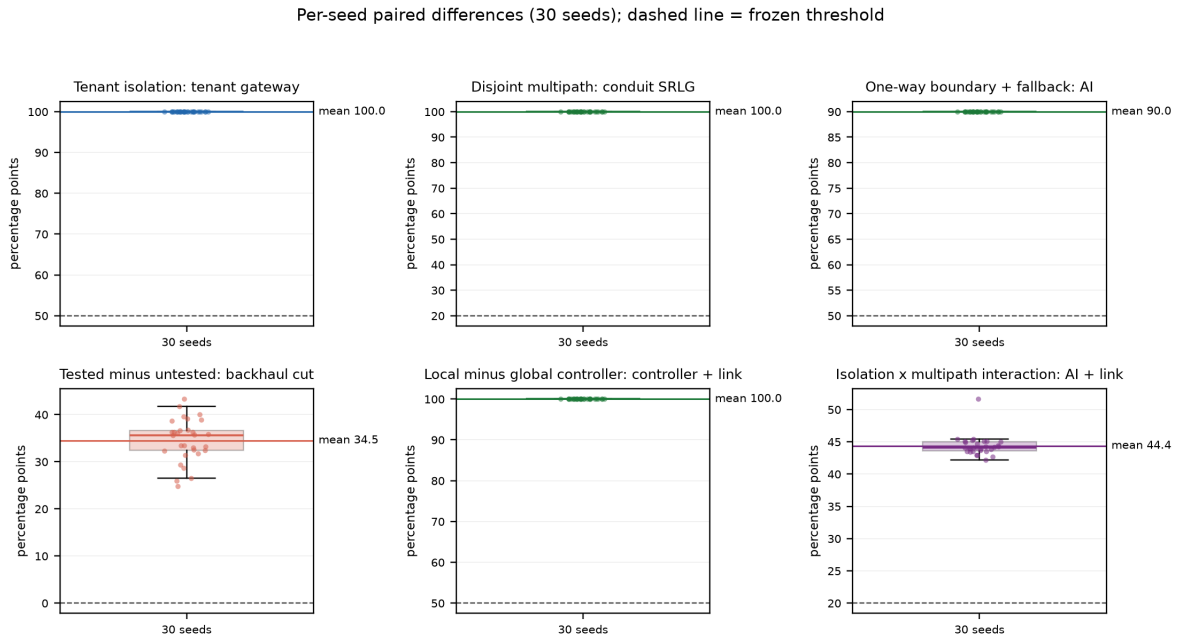


Figure 4. Per-seed paired differences for six mechanism comparisons over 30 paired seeds. Boxes show the interquartile range, points are individual seeds, the solid line is the mean and the dashed line is the frozen threshold. Data: `docs/paper-tables/common-mode-per-seed.csv`; generated by `tools/make_common_mode_figures.py`.

5.10 Sensitivity and break-even for the stipulated constants

Every number above depends on a stipulated constant, so the interesting question is not only whether a check passes at the stipulated value but how far the value could move before the check changes. We sweep each constant one at a time over 25 values, with the other constants, topology, events and 30 seeds fixed, and locate the nearest value at which each criterion stops passing. The grid and the full per-criterion results are committed in `docs/paper-tables/common-mode-sensitivity.csv` and `docs/paper-tables/common-mode-break-even.csv`; the table below is pasted from the generated Markdown, not retyped.

Table 5. Break-even values for the stipulated constants. Grid: 25 values per constant, 30 paired seeds, linear interpolation between adjacent grid points. A break-even is the value at which the named criterion stops passing; "none in range" means it passed at every swept value.

Constant	Stipulated	Sweep	Primary break-even	Claim that fails there	Margin
one-way local-policy efficiency	0.9 fraction	0.5-1 fraction	0.85	H18: normal-operation cost of the one-way boundary <= 15 points	-0.05
tested fallback activation probability	0.98 fraction	0-1 fraction	0.222334	tested fallback beats untested under the backhaul cut (gap > 0)	-0.757666
untested fallback activation probability	0.6 fraction	0-1 fraction	none in range	criterion survives the whole swept range	

Constant	Stipulated	Sweep	Primary break-even	Claim that fails there	Margin
tested fallback activation delay	2 s	0-25 s	18.166	tested fallback beats untested under the backhaul cut (gap > 0)	+16.166
untested fallback activation delay	8 s	0-30 s	none in range	criterion survives the whole swept range	
tested fallback capacity	0.5 Gbps	0-1 Gbps	0.12172	tested fallback beats untested under the backhaul cut (gap > 0)	-0.37828
untested fallback capacity	0.3 Gbps	0-1 Gbps	none in range	criterion survives the whole swept range	

For H18, the binding clause is the normal-operation cost (0.85), not the absolute 80% level (0.80); the model check fails at the higher of the two values. The tested fallback is far from its break-even on probability, delay and capacity: the stipulated 0.98 activation probability would have to fall to 0.22, the 2 s delay would have to rise to 18.2 s, or the 0.50 Gbps capacity would have to fall to 0.122 Gbps before it stopped beating the untested fallback under the backhaul cut. The untested fallback would have to improve beyond every value in the swept range to erase the 34.5-point gap. That is a robustness statement about the model, not a confidence interval about a production deployment.

Capacity is the one two-sided case. H19's interaction term is `A6 - A9` under the combined failure: if the tested fallback becomes too capable, A9 alone covers more of the event and the interaction term falls below 20. The H19 interaction break-even is 0.767 Gbps, while H19's absolute A6 level still fails only below 0.80 efficiency. The H19 check is a claim about super-additivity, not a claim that a stronger fallback is worse; the paper reports both break-evens rather than choosing the flattering one.

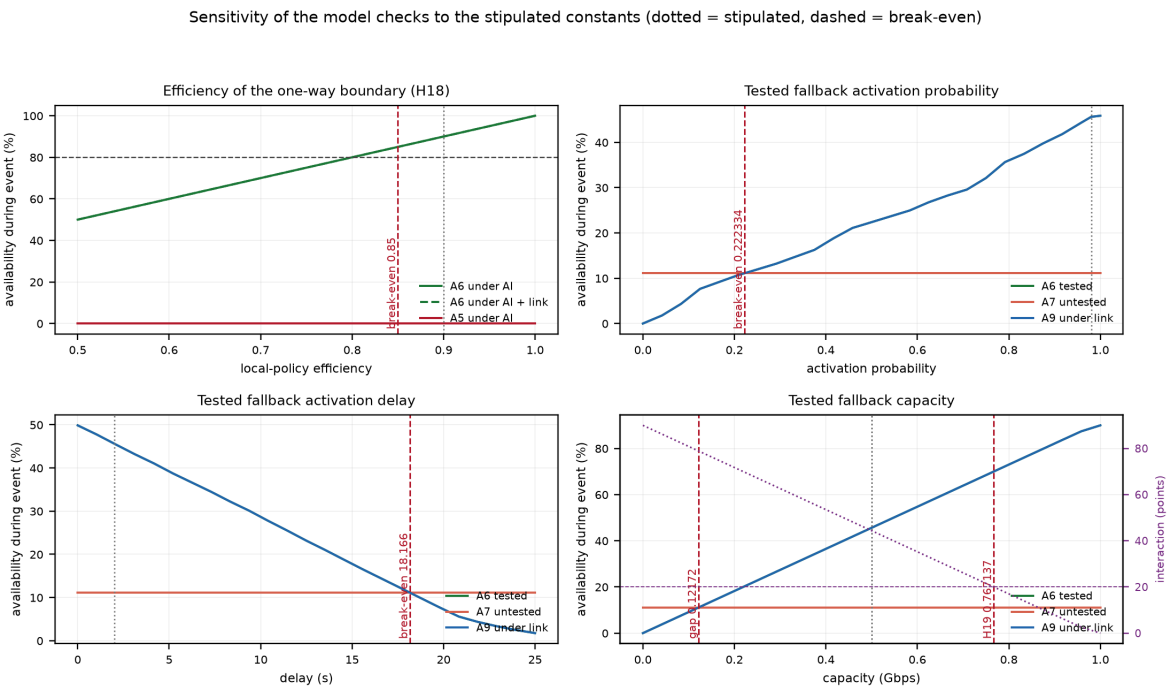


Figure 5. Sensitivity of the model checks to the stipulated constants. Each panel varies one constant; the dotted vertical line is the stipulated value, the dashed red line is the nearest break-even, and the horizontal dashed lines are the frozen thresholds. The top-right panel varies the tested-fallback activation probability; the bottom panels vary its delay and capacity. Data: `docs/paper-tables/common-mode-sensitivity.csv` and `common-mode-break-even.csv`; generated by `tools/make_common_mode_figures.py`.

6. Discussion

6.1 What the comparison shows

Under the stipulated model, the comparison is consistent with three claims.

First, **current cloud isolation is the wrong cut for common-mode failure.** Tenant isolation, confidential computing, regions and provider diversity each remove a real failure mode, and the model check reproduces that they do: A4 fixes the tenant gateway, A3 fixes the conduit, A5 fixes identity. But none of them cuts the shared AI/control-plane dependency. The result is not an argument against tenant isolation; it is an argument against treating tenant isolation as sufficient for a provider-level dependency.

Second, **multipath must be independent, not merely multiple.** A2 and A3 have the same path count and opposite outcomes under the conduit failure. The path independence index is a better predictor than the number of paths. A path set that shares a control plane can even amplify a correlated failure, because every path depends on the same identity or AI endpoint.

Third, **the controller is a dependency.** A8 behaves like A5 until its controller fails, then loses the multipath advantage exactly when it is needed. A global cooperative controller is a new common-mode carrier; locality is a design variable. This is the cooperative-backhaul form of the concentration problem: the mechanism that makes self-scaling possible is also the mechanism that can centralize it.

6.2 Design implications

The model checks translate into five design rules, conditional on the scenario.

1. **Inventory shared dependencies, then cut the control plane, not only the tenant boundary.** A tenant-isolation review that does not include identity, DNS, the AI endpoint and the orchestrator is incomplete.
2. **Measure path independence, not path count.** Report SRLG membership, provider control-plane membership and shared control-service membership for every path set. A path set with one SRLG is one path.
3. **Treat the fallback as a tested state.** Record activation probability, delay and capacity. An untested fallback is a configuration, not a mitigation; in the model it was worth 34.5 points less than a tested one.
4. **Keep cooperative control local or partition-tolerant.** A global controller can be part of the data plane's efficiency story, but it must not be on the availability path. If it is, the controller is a single point of failure with a scheduling interface.
5. **Budget the normal-operation cost of isolation.** The one-way boundary cost 10 points in this model. That is the price of local policy; it should be stated, not hidden.

6.3 The A9/A6 trade and the meaning of "both"

The cost-resilience comparison is deliberately not a single ranking. A9 (isolation only) matches A6 on the common-mode mean at less than half the cost, because both use the same boundary and fallback. A6 is better only when an independent link failure is also in scope (90% vs 45.6%). This is a useful negative result for the strongest form of the "both axes" thesis: isolation plus a tested fallback can be sufficient for a common-mode-only failure model. The argument for multipath is conditional on independent path failures being part of the threat model. The paper's claim is therefore narrower and more defensible: the two axes are independent mitigations, and the right mix depends on the failure mix, not on a universal ranking.

6.4 Relation to the motivating outages

The 13 November 2026 events are used as motivating examples, not as evidence, and they are kept separate. The FAA reported a nationwide ground stop after an old telecommunications system carrying air-traffic control communications was overloaded; the stop lasted about four hours and twenty minutes, and the FAA and the Department of Homeland Security reported no evidence of a cyber or AI attack [15,16]. Microsoft reported a Microsoft 365 outage attributed to an issue introduced during an earlier network capacity upgrade, affecting Outlook, Teams, Copilot and other services, and said a full post-incident report would follow [17,18]. ChatGPT and Claude also reported disruption during the same period [18], but the public reports do not establish a common cause with the Microsoft outage. The 2-3 September 2026 events are a second, separate coincidence: Microsoft 365 degradation [19] and elevated errors at ChatGPT, Claude and Grok [20,21] on the same day, without a shared root cause in the public reports.

The structural comparison does not depend on either root cause, and the paper does not claim that the model explains these events. What the testbed adds is a controlled demonstration of the mechanism: a shared dependency, a boundary that does not cut it, and a path set that does not survive it. The empirical test is the separate postmortem study; the out-of-sample check is to code the eventual Microsoft post-incident report and the AI providers' reports with the same codebook, without adding them to the analysis set.

7. Threats to Validity

Construct validity. Availability is a capacity/dependency calculation, not a packet-level measurement. It ignores queueing, congestion control, TCP retransmission, reordering and application behaviour. The 0.9 local-policy efficiency, the activation probabilities, the delays and the 0.5/0.3 fallback capacities are stipulated. The cost index is an accounting device, not a price. The path independence index treats SRLG and provider identity as the only independence dimensions; a real deployment may have more.

Stipulated constants. Section 5.10 turns each stipulated constant into a break-even value. The exercise is one-factor-at-a-time: it holds the remaining constants and the topology fixed, so it does not cover interactions between constants. The break-even values are linearly interpolated between 25 grid points, not solved analytically. They show how far a verdict is from the stipulated operating point; they do not give a confidence interval, and a combination of changes could cross a boundary that no single change crosses.

Internal validity. The model is deterministic given a seed, and the 30 seeds vary event timing, demand and fallback draws, not the topology. All five checks pass with large margins because the arms isolate the mechanisms by design. This is a demonstration under a frozen scenario, not a statistical estimate of an effect size. The arms differ only in mechanism and share the same draws, which rules out a seed confound; it does not rule out a scenario confound. No constant was changed after the freeze.

External validity. The scenario has six tenants, three aggregation sites, three providers and two clouds. It is not a model of a specific cloud, and the current-cloud mapping in Table 1 is a structural analogy. Different SRLG groupings, control-plane layouts or demand levels could change the levels. The qualitative ordering is the claim; the exact percentages are scenario-specific.

No empirical calibration. The model is not fitted to the postmortems. The empirical question is tested separately, by the companion postmortem study with its own frozen codebook and coders. An agreement between the two would be support for the model's relevance, not for its mechanism; a disagreement would be a finding about the model, not a reason to retune it.

Conclusion validity. The hypotheses were frozen before the simulator existed, and the verdict rules are in the benchmark. A refutation would have been reported. The all-pass outcome is a limitation as well as a result: the scenario was designed to make the mechanisms observable, so it is not a random sample of architectures. The checks are proofs about the model, not about the world; the paper's claims are conditional on the stipulated scenario. The real-stack extension is the next fidelity and is not claimed here.

Security. The data diode is modelled as an availability cut, not as a cryptographic control. No key management, authentication strength or side-channel behaviour is evaluated. The one-way boundary's security properties are out of scope.

8. Conclusion

Concentration is the risk. A shared dependency with no independent failure domain behind it turns a local fault into a correlated outage, and the modern cloud stack is as capable of this as the legacy stack. Isolation and multipath are the two independent mitigations. Isolation stops a fault at a boundary; multipath keeps service running across independent failure domains. Neither substitutes for the other, and a tested fallback is the condition that makes either one real.

Under the frozen scenario the five model checks pass. Tenant isolation fixes the tenant gateway but not identity or AI; three paths over one conduit fail together; a one-way boundary with a tested fallback holds 90% under an AI failure; the isolation-multipath interaction is +44.4 points; and a global cooperative controller loses 100 points when it fails. The comparison with stylized current-cloud architectures shows the gap: current cloud improves along one axis at a time, while no current-cloud arm reaches 90% under a combined AI and link failure. The qualification is equally important: isolation plus a tested fallback can match the combined architecture on common-mode events at lower cost, so multipath is justified by independent path failures, not by common-mode failure alone.

The sensitivity analysis is part of the result, not an appendix. It says that H18 would fail only below 0.85 efficiency, that the tested-fallback gap is far from its break-even on probability, delay and capacity, and that H19's super-additivity has a two-sided capacity boundary at 0.767 Gbps. Those are statements about the model's stipulated operating point. The empirical question is separate and is not answered here: the companion postmortem study codes provider reports under its own frozen codebook, and the eventual Microsoft and AI-provider post-incident reports are the out-of-sample check. The next fidelity for the model is a real-stack variant on Linux network namespaces: two disjoint paths, ECMP, link-down and `tc` outages, and an nftables control-plane cut, compared against this flow-level reference. The structural result is a map of which cuts and which paths matter; the real-stack test is how to find out what the map costs in packets.

References

1. P. Mell and T. Grance, "The NIST Definition of Cloud Computing," NIST Special Publication 800-145, 2011.
2. Amazon Web Services, "Regions and Availability Zones," AWS Documentation. [Online]. Available: AWS documentation
3. Microsoft, "Availability zones overview," Azure Reliability Documentation. [Online]. Available: Azure documentation
4. Google Cloud, "Regions and zones," Compute Engine Documentation. [Online]. Available: Google Cloud documentation
5. A. Avizienis, J.-C. Laprie, B. Randell and C. Landwehr, "Basic concepts and taxonomy of dependable and secure computing," *IEEE Transactions on Dependable and Secure Computing*, vol. 1, no. 1, pp. 11-33, 2004.
6. H. S. Gunawi, M. Hao, T. Leesatapornwongsa, T. Patana-anake, T. Do, J. Adityatama, K. J. Eliazar, A. Laksono, J. F. Lukman, V. Martin and A. D. Satria, "What Bugs Live in the Cloud? A Study of 3000+ Issues in Cloud Systems," in *Proceedings of the ACM Symposium on Cloud Computing (SoCC)*, 2014.
7. A. Basiri, N. Behnam, R. de Rooij, L. Hochstein, L. Kosewski, J. Reynolds and C. Rosenthal, "Chaos Engineering," *IEEE Software*, vol. 33, no. 3, pp. 35-41, 2016.
8. National Institute of Standards and Technology, "Guide to Operational Technology (OT) Security," NIST Special Publication 800-82 Revision 3, 2023.
9. Siemens Mobility, "Protection of OT systems," and "CoreShield Data Capture Unit." [Online]. Available: Siemens Mobility and CoreShield DCU
10. A. Ford, C. Raiciu, M. Handley, O. Bonaventure and C. Paasch, "TCP Extensions for Multipath Operation with Multiple Addresses," RFC 8684, 2020.
11. C. Hopps, "Analysis of an Equal-Cost Multi-Path Algorithm," RFC 2992, 2000.
12. K. Kompella and Y. Rekhter, "Routing Extensions in Support of Generalized Multi-Protocol

Label Switching (GMPLS),” RFC 4202, 2005.
13. M. Nikolic, “Which Rewrites Survive? Pre-registered Tests of a Field-Class Model of Encapsulation, Middleboxes and Decapsulation on Linux,” repository paper, 2026. [Online]. Available: Paper A
14. M. Nikolic, “Why Fair-Queueing Beats the Fick Principle,” repository paper, 2026. [Online]. Available: Paper B
15. Bernama-Anadolu, “Nationwide Ground Stop Issued At US Airports Over ‘Communication Issue’: Transport Secretary,” 14 November 2026. [Online]. Available: Bernama
16. Yonhap News Agency, “US-wide flight communications failure causes a four-and-a-half-hour all-stop; fallout continues” (in Korean; English title supplied for readability), 14 November 2026. [Online]. Available: Yonhap
17. Cambridge Independent / PA News, “Microsoft ‘deeply sorry’ for disruption caused by major global outage,” 13 November 2026. [Online]. Available: Cambridge Independent
18. Kurier, “Ursache von Microsoft-Störung war Netzwerk-Upgrade,” 13 November 2026. [Online]. Available: Kurier
19. Android Authority, “Is Microsoft Outlook, OneDrive, and more down for you? Here’s what’s going on (Update),” 2-3 September 2026. [Online]. Available: Android Authority
20. APA, “AI outage hits Claude, ChatGPT and Grok,” 3 September 2026. [Online]. Available: APA
21. GIGAZINE, “ChatGPT, Claude, and Grok all went down at almost the same time,” 4 September 2026. [Online]. Available: GIGAZINE
22. H. S. Gunawi, M. Hao, R. O. Suminto, A. Laksono, A. D. Satria, J. Adityatama and K. J. Eliazar, “Why Does the Cloud Stop Computing? Lessons from Hundreds of Service Outages,” in *Proceedings of the ACM Symposium on Cloud Computing (SoCC)*, 2016, DOI 10.1145/2987550.2987583.
23. S. Talluri, D. Niewenhuis, X. Chu, J. Kyselica, M. Cetin, A. Balgavy and A. Iosup, “Cloud Uptime Archive: Open-Access Availability Data of Web, Cloud, and Gaming Services,” arXiv:2504.09476, 2025. [Online]. Available: arXiv
24. P. S. L. Paladugu, “How Configuration and Deployment Changes Trigger Large Cloud and SaaS Outages: An Empirical Study of 70 Public Postmortems,” Zenodo preprint, 22 October 2026, DOI 10.5281/zenodo.23585116. [Online]. Available: Zenodo
25. Amazon Web Services, “Post-Event Summaries.” [Online]. Available: AWS
26. Microsoft, “Azure status history: Post Incident Reviews.” [Online]. Available: Azure
27. Google Cloud, “Service Health.” [Online]. Available: Google Cloud
28. Cloudflare, “Post Mortem” blog archive. [Online]. Available: Cloudflare
29. L. A. Barroso, J. Clidaras and U. Hölzle, *The Datacenter as a Computer: Designing Warehouse-Scale Machines*, 3rd ed. Morgan and Claypool, 2018.
30. National Institute of Standards and Technology, “Hardware-Enabled Security for Server Platforms: Security Architecture and Design,” NIST IR 8320, 2021.

Appendix: Artifact Index

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Results and deviations	`docs/common-mode-results.md`
Testbed guide	`docs/common-mode-testbed.md`
Simulator	`simulation/common_mode.py`

Artifact	Path
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Seed and sensitivity figures	figures/common_mode_seed_distributions.png, figures/common_mode_sensitivity.png